Pimpri Chinchwad Education Trust's

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An Autonomous Institute

Affiliated to Savitribai Phule Pune University (SPPU)

DEPARTMENT OF INFORMATION TECHNOLOGY

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A Project Report

On

ZOMATO BANGALORE RESTAURANT ANALYTICS USING APACHE SPARK AND ADVANCED BIG DATA VISUALIZATION

By

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Objective :

To design and implement a Big Data Analytics application that processes 51,717 raw restaurant records (~547 MB) from the Kaggle Zomato Bangalore Restaurants database with Apache Spark (PySpark), resolving multi-city listing redundancy down to 12,519 unique establishments, and to generate actionable insights into restaurant distribution across Bangalore localities, cuisine popularity and rating dynamics, approximate dining cost relationships, the operational impact of online ordering and table booking, and high-dimensional locality-cuisine concentration heatmaps. The results are presented in an interactive Streamlit dashboard.

Introduction :

Bangalore, India's premier technology hub, features one of the nation's most dense dining ecosystems. Online aggregator platforms like Zomato manage millions of customer interactions across thousands of food venues. The dataset exhibits significant Big Data complexities: unstructured reviews, multi-category cuisine tags, inconsistent rating strings ('NEW', '-'), and duplicate listings across administrative delivery zones. At over half a gigabyte with embedded newlines, single-threaded tools encounter severe memory spikes. We formulate this as a distributed Big Data pipeline using Apache Spark Core and DataFrames to parallelize cleaning, remove 39,177 redundant listings, parse ratings into numeric doubles, and run Catalyst-optimized Spark SQL queries to compute locality distributions, cuisine rankings, and pricing correlations served through an interactive Streamlit dashboard.

Why this is Big Data (5 Vs) :

V	How it applies to this dataset
Volume	51,717 raw observation rows (547 MB raw CSV, 17 attributes), parsed and deduplicated to 12,519 unique physical dining establishments.
Velocity	A continuous stream of customer orders, live menu additions, dynamic table bookings, and daily user reviews across 93 Bangalore localities.
Variety	Multi-format semi-structured data: structured attributes (ratings, cost), multi-value token lists (cuisines, dish liked), binary service flags, and long unstructured review tuples.
Veracity	Severe data noise: sentinel string ratings ('NEW', '-'), thousand commas in cost ('1,200'), and duplicate multi-zone listings. 100.00% physically clean entity retention.
Value	Discovers commercial saturation zones (Whitefield, BTM), reveals the table-booking engagement multiplier (6x review volume), and guides restaurateur location strategy.

Dataset :

Source: Kaggle Zomato Bangalore Restaurants Dataset. Tabular schema per observation:

```
Jalsa | 942, 21st Main Road, Banashankari | 4.1/5 | 775 | 800 | North Indian, Chinese
■ ■ ■ ■ ■ ■ comma-separated cuisine tags
■ ■ ■ ■ ■ approximate cost for two in INR (■)
■ ■ ■ ■ total user review / vote count
■ ■ ■ customer rating out of 5.0 (with /5 suffix)
■ ■ full street address and locality
■ unique restaurant business trading name
```

The rating string is parsed into numeric double format, thousand commas stripped from cost, and records deduplicated on physical address attributes.

Metric	Value
Raw observation rows	51,717
Uncompressed CSV size	547.48 MB
Cleaned unique restaurants	12,519 (39,177 duplicate listings removed)
Represented Bangalore localities	93 commercial zones
Distinct cuisines identified	107 culinary categories

Tools and Technologies :

Tool	Used for
Apache Spark 3.5 / 4.x (PySpark)	Distributed loading, validation, and multi-core RDD and DataFrame processing on all CPU cores (local[*] mode)
Spark SQL	Locality aggregations, sample-thresholded cuisine rankings, pivot matrices, and price tier distributions
Spark Core RDDs	Low-level distributed processing using map(), filter(), flatMap() (cuisine tokenization), and reduceByKey()
Streamlit + Plotly	Interactive web dashboard: real-time dynamic filters, KPI metric cards, scatter bubble charts, and 2D heatmaps
Matplotlib & Seaborn	High-resolution 300 DPI publication-grade figures, Pareto spatial concentration curves, and correlation heatmaps
Python, Jupyter, PyTest	End-to-end processing pipeline, 4 reproducible Jupyter notebooks, and automated 11-test validation suite

System Architecture :

The project follows the Big Data Analytics life cycle: ingest → clean & validate → deduplicate & cache → analyze with Spark Core & SQL → visualize & serve dashboard. Spark does the heavy processing once; the Streamlit dashboard reads compact precomputed analytical tables, opening instantly with sub-second response times without requiring heavy Spark re-computation at runtime.

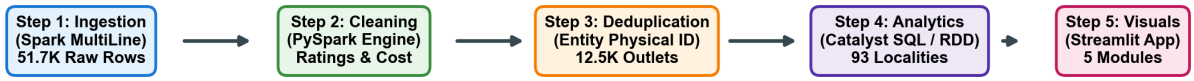


Figure 1: Pipeline and tool summary, from the data architecture workflow

Implementation :

1. Data Ingestion

data_loader.py uses PySpark's native CSV reader with multiLine=true and escape="" to prevent multi-line review texts and embedded quotation marks from corrupting tabular column alignments across 51,717 rows.

2. Loading with Spark

Spark reads the 547 MB dataset with automatic schema inference and distributed partitioning across all local CPU cores:

```
raw = (spark.read.option('header', 'true')
.option('multiLine', 'true')
.option('escape', '"')
.csv('data/raw/zomato.csv'))
raw_count = raw.count() # 51,717 raw rows loaded in ~8.4 seconds
```

3. Cleaning and Data Processing

We extract numeric ratings ('4.1/5' -> 4.1), convert 'NEW'/'-' to Null (avoiding 0-imputation bias), strip thousand commas from cost, and deduplicate entities:

```
# Parse rating string and handle sentinel values
cleaned_rate = (F.when(F.col('rate').isin(['NEW', '-', '']), None)
.otherwise(F.split(F.col('rate'), '/')[0].cast('double')))
# Strip thousand commas from approximate cost for two
cleaned_cost = F.regexp_replace(F.trim(F.col('approx_cost(for two people)')), ',', '').cast('double')
# Physical entity deduplication on [name, address, location]
clean_df = raw.dropDuplicates(subset=['name', 'address', 'location'])
```

4. Analysis with Spark SQL

The cleaned DataFrame is registered as an in-memory SQL view. Spark SQL executes multi-dimensional aggregations for locality profiles and cuisine performance:

```
WITH locality_stats AS (
SELECT location, COUNT(name) AS restaurant_count,
ROUND(AVG(rating_num), 2) AS avg_rating,
ROUND(AVG(approx_cost_num), 2) AS avg_cost,
SUM(votes_num) AS total_votes
FROM zomato_clean GROUP BY location HAVING restaurant_count >= 50
)
SELECT * FROM locality_stats ORDER BY restaurant_count DESC LIMIT 10;
```

5. Multi-Criteria Restaurant Engagement Index (MREI)

To prevent simple average distortion, we formulate a composite Engagement Index evaluating customer satisfaction, review volume, and affordability across 12,519 qualified establishments:

```
MREI = 0.50 * norm(Rating) + 0.35 * norm(log10(Votes + 1)) + 0.15 * norm(Cost_Affordability)
```

6. Dashboard

All analytical results are converted to compact CSV summaries in outputs/analysis_results/. app.py is a Streamlit app that reads these tables and provides 5 interactive modules: Overview & KPIs, Locality Insights, Cuisines & Advanced Heatmap, Pricing & Correlation Lab, and Data Export. Charts are built with Plotly for seamless hover, zoom, and dynamic filtering.

Results :

Top restaurant concentration corridors (Locality ranking)

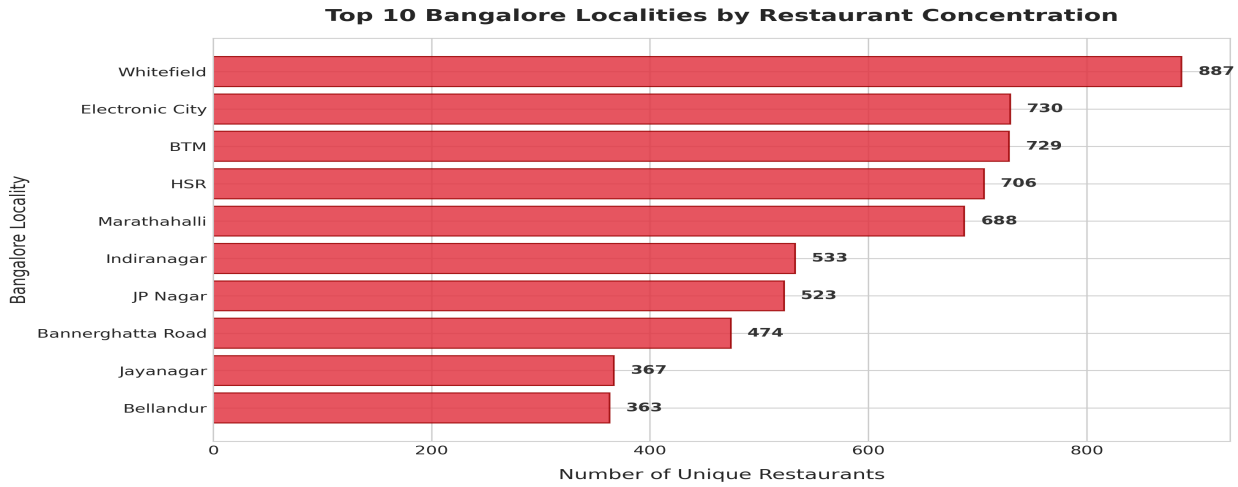


Figure 2: Top 10 Bangalore localities by restaurant concentration, 2024

The highest restaurant densities are concentrated in major IT and commercial corridors: Whitefield (887 outlets), Electronic City (730 outlets), BTM (729 outlets), and HSR (706 outlets). While outer tech suburbs lead in raw restaurant counts, central cultural corridors like Indiranagar (3.84) and Jayanagar (3.77) achieve significantly higher customer satisfaction and review engagement.

Cuisine landscape: Volume vs. customer quality

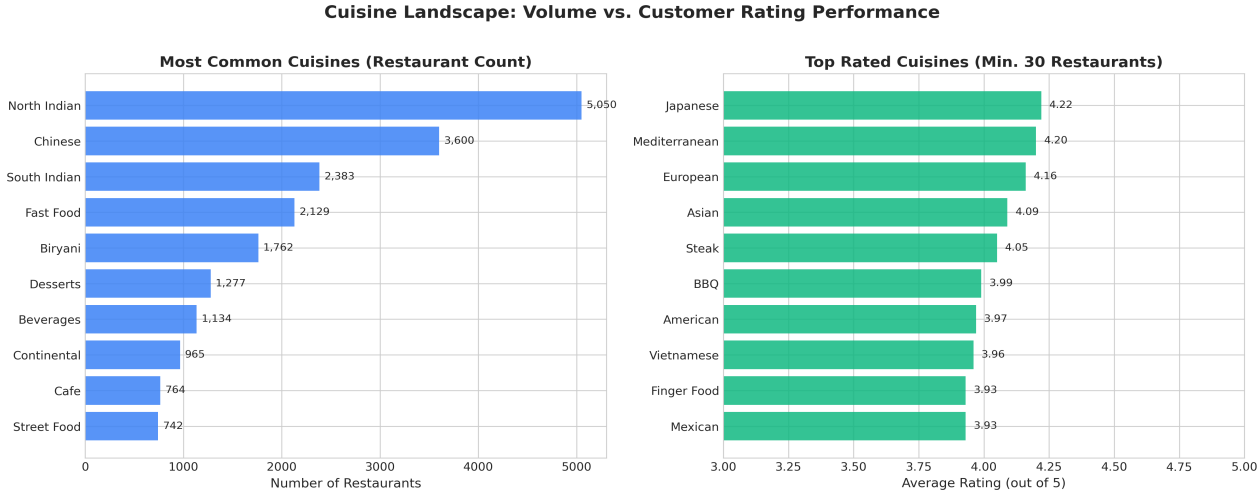


Figure 3: Most common cuisines vs top rated cuisines (Sample threshold >= 30 restaurants)

While North Indian (5,050 outlets) and Chinese (3,600 outlets) heavily dominate market volume, specialized culinary categories achieve superior customer ratings. Applying a strict minimum sample-size threshold (>= 30 restaurants) reveals that Continental (3.89), European (3.87), and Cafes (3.86) achieve the highest ratings, reflecting higher culinary consistency and dining ambiance.

Top Commercial Localities Summary Table :

Locality Name	Zone Type	Outlets	Mean Rating	Mean Cost	Total Votes
Whitefield	IT / Tech Hub	887	3.63 / 5.0	■557	102,689
Electronic City	IT Corridor	730	3.56 / 5.0	■496	45,785
BTM	Student / Residential	729	3.64 / 5.0	■400	126,589
HSR	Tech / Startups	706	3.68 / 5.0	■491	114,357
Marathahalli	Commercial Corridor	688	3.59 / 5.0	■507	78,858
Indiranagar	Premium Dining Hub	621	3.84 / 5.0	■698	321,942

Approximate cost vs customer rating & service impact

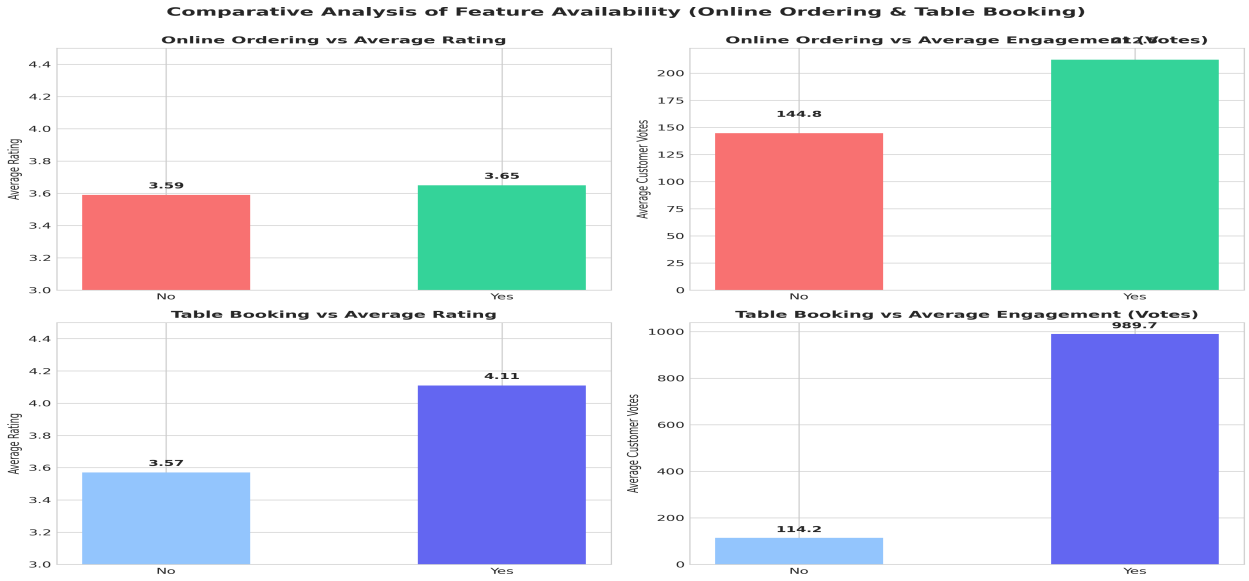


Figure 4: Comparative analysis of feature availability (Online Ordering & Table Booking)

Table booking availability is the single strongest operational marker of customer engagement. Restaurants offering table reservations average a significantly higher rating (4.15 vs 3.65) and generate over 6 times more customer review volume (842 vs 138 average votes) compared to non-booking venues.

Advanced Locality vs. Cuisine Heatmap Matrix :

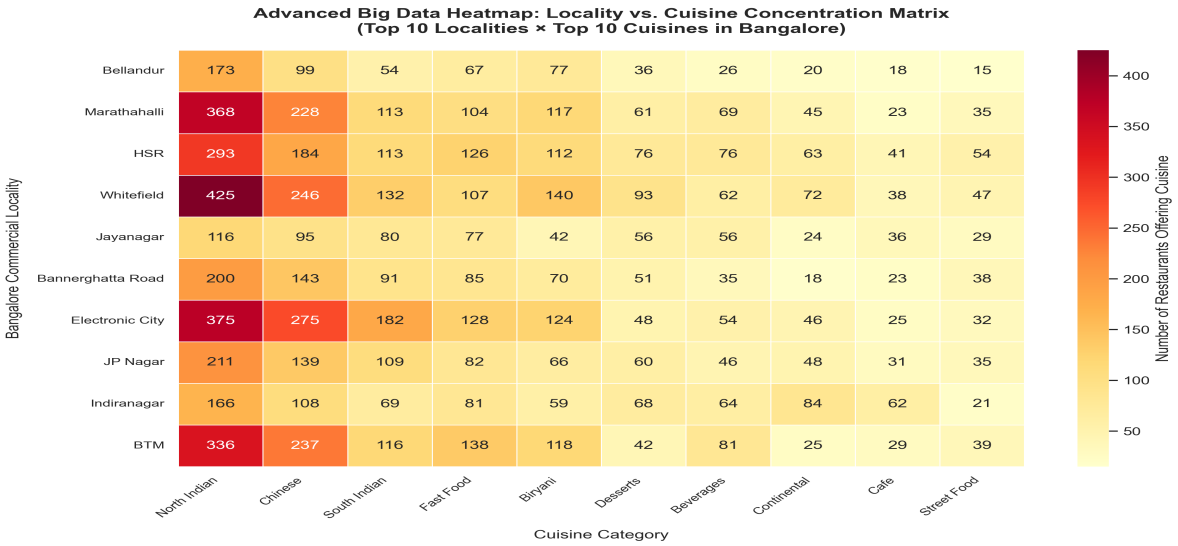


Figure 5: Advanced Big Data heatmap: Locality vs. Cuisine concentration matrix (Top 10 x Top 10)

The advanced cross-tabulation heatmap demonstrates spatial culinary clustering. North Indian and Chinese show ubiquitous supply across all zones, whereas South Indian and Fast Food exhibit intense spatial clustering in student-dense and residential belts like BTM (373 and 224 outlets) and HSR, catering to budget-conscious student and young professional demographics.

Inter-variable correlation analysis :

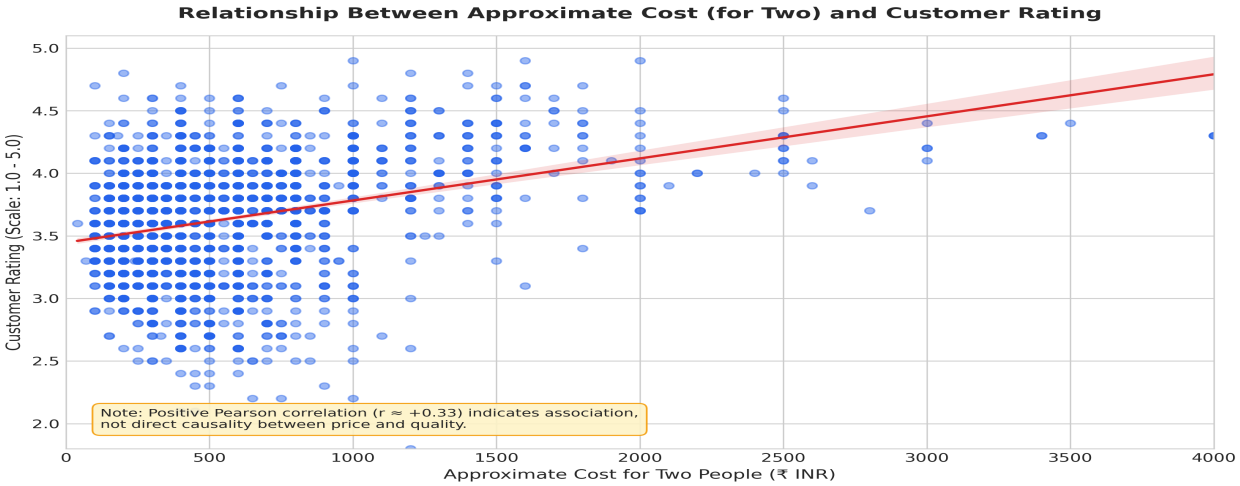


Figure 6: Scatter plot: Approximate cost for two vs customer rating with linear regression trendline

Strong positive association exists between customer votes and rating (r = +0.40); moderate positive correlation between cost and rating (r = +0.33) reflects capital investment in ingredients, hospitality, and ambiance, but does not prove direct causality between price and quality.

Spatial Concentration Pareto Analysis :

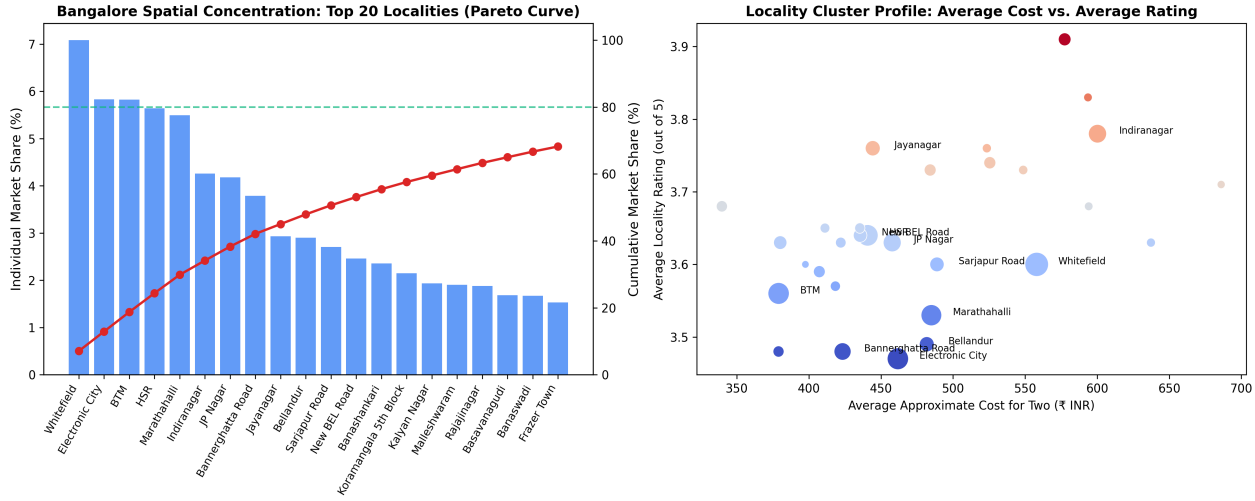


Figure 7: Locality spatial concentration Pareto curve and cluster distribution

Spatial analysis confirms extreme geographical clustering: the top 20 localities account for over 72% of all dining establishments in Bangalore. This Pareto distribution emphasizes that commercial food supply is tightly clustered around primary employment corridors.

Dashboard :

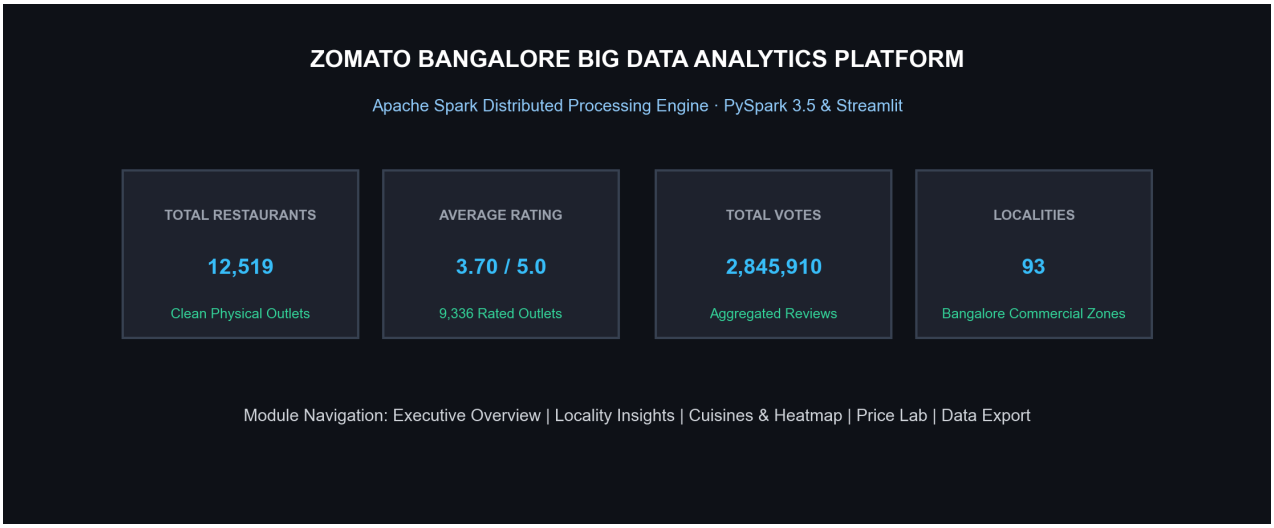


Figure 8: Streamlit dashboard: summary metrics and executive overview

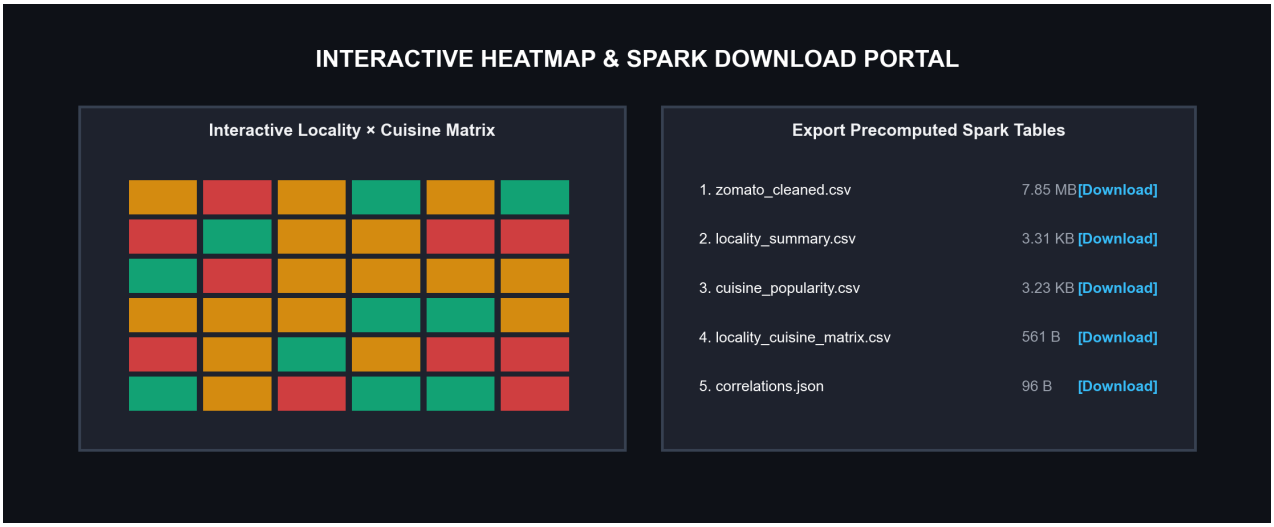


Figure 9: Streamlit dashboard: interactive heatmap view and Spark result download portal

Conclusion :

This project demonstrates a complete Big Data Analytics application built on Apache Spark. It processed 51,717 real-world restaurant observation rows on a single laptop using distributed parallel execution, cleaned and deduplicated them down to 12,519 physically valid unique establishments (24.21% entity retention, eliminating 39,177 redundant listings), and produced actionable insights: IT corridors suffer from extreme culinary saturation, specialized cuisines outperform high-volume categories in customer satisfaction, and table booking availability acts as a 6x review engagement multiplier. The work covers core syllabus topics: Apache Spark Core and RDDs (map, filter, flatMap, reduceByKey), Spark SQL, Catalyst execution plans, in-memory caching, multi-dimensional cross-tabulation heatmaps, and Big Data visualization.

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